

Module - 6 : Project management

Structure of projects, phases of project management - planning, scheduling, controlling phase, work breakdown structure, project control charts, network planning, PERT & CPM, Network components & precedence relationships, Critical path analysis, probability in PERT analysis, Theory of crashing (Theory Only), Theory of Constraints (Theory only).

I. MEANING : PERT / CPM is a technique used for assisting project managers carry out their responsibilities. This technique has proved to be useful for all jobs or projects that have an element of uncertainty in the estimation of duration, as is the case with new types of projects the likes of which have never been taken up before.

PERT (*Programme Evaluation and Review Technique*) was developed in 1956–58 by a research team to help in the planning and scheduling of the US Navy's Polaris Nuclear Submarine Missile project involving thousands of activities. The objective of the team was to efficiently plan and develop the Polaris missile system. This technique has proved to be useful for projects that have an element of uncertainty in the estimation of activity duration, as is the case with new types of projects which have never been taken up before.

CPM (*Critical Path Method*) was developed by E.I. DuPont company along with Remington Rand Corporation almost at the same time, 1956-58. The objective of the company was to develop a technique to monitor the maintenance of its chemical plants. This technique has proved to be useful for developing time-cost trade-off for projects that involve activities of repetitive nature.

II. SIGNIFICANCE OF USING PERT/CPM

1. A network diagram helps to translate complex project into a set of simple and logical arranged activities and therefore,
 - Helps in the clarity of thoughts and actions.
 - Helps in clear and unambiguous communication developing from top to bottom and vice versa among the people responsible for executing the project.
2. Detailed analysis of a network helps project in charge to peep into the future because
 - Difficulties and problems that can be reasonably expected to crop up during the course of execution, can be foreseen well ahead of its actual execution.
 - Delays and holdups during course of execution are minimized. Corrective action can also be taken well in time.
3. Isolates activities that control the project completion and therefore, results in expeditious completion of the project.
4. Helps in the division of responsibilities and therefore, enhance effective coordination among different departments/agencies involved.
5. Helps in timely allocation of resources to various activities in order to achieve optimal utilization of resources.

III. PHASES OF PROJECT MANAGEMENT

In general, project management consists of three phases: Planning, Scheduling and Control.

1. Project planning phase : In order to understand the sequencing or precedence relationship among activities in a project, it is essential to draw a network diagram. The steps involved during this phase are listed below:

- Identify various activities (tasks or work packages/elements) to be performed in the project, that is, develop a breakdown structure (WBS).
- Determine the requirement of resources such as men, materials, machines, money, etc., for carrying out activities listed above.
- Assign responsibility for each work package. The work packages corresponds to the smallest work efforts defined in a project and forms the set of tasks that are the basis for planning, scheduling and controlling the project.
- Allocate resources to work packages.
- Estimate cost and time at various levels of project completion.
- Develop work performance criteria.
- Establish control channels for project personnel.

2. Scheduling phase : Once all activities have been identified and given unique codes, the project scheduling (when each of the activities is required to be performed) is taken up. Prepare an estimate of the likelihood of the project to be completed on or before the specified time. The steps involved during this phase are listed below:

- Identify all people who will be responsible for each task.
- Estimate the expected duration(s) of each activity, taking into consideration the resources required for their execution in the most economic manner.
- Specify the interrelationship (i.e. precedence relationship) among various activities.
- Develop a network diagram, showing the sequential interrelationship between various activities.
- For this, tips such as; *what* is required to be done; *why* it must be done, can it be dispensed with; *how* to carry out the job; *what* must precede it; *what* has to follow; *what* can be done concurrently, may be followed.
- Based on these time estimates, calculate the total project duration, identify critical path; calculate floats; carry out resources smoothing (or levelling) exercise for critical (or scarce) resources, taking into account the resource constraints (if any).

3. Project control phase : Project control refers to the evaluation of the actual progress (status) against the plan. If significant differences are observed, then remedial (modifying planning) or reallocation of resources measures are adopted in order to update and revise the uncompleted part of the project.

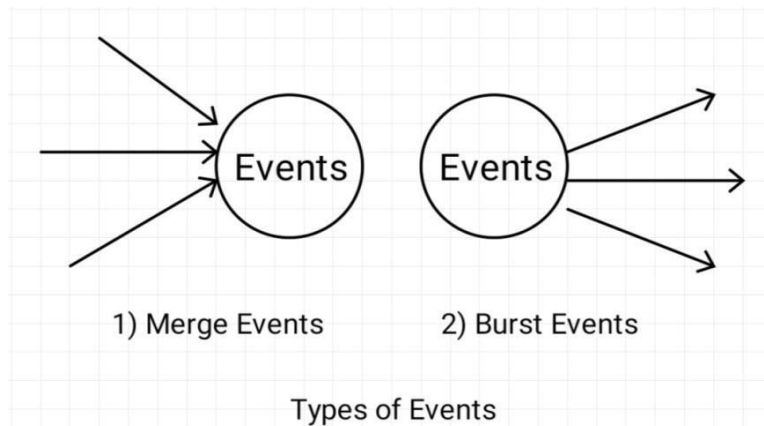
IV. PERT/CPM NETWORK COMPONENTS/ELEMENTS AND PRECEDENCE RELATIONSHIPS

PERT/CPM network consists of two major components.

1. Events : Events in the network diagram represent project milestones, such as the start or the completion of an activity (task) or activities, and occur at a particular instant of time at which some specific part of the project has been or is to be achieved. Events are commonly represented by circles (nodes) in the network diagram.

The events can be further classified into the following two categories:

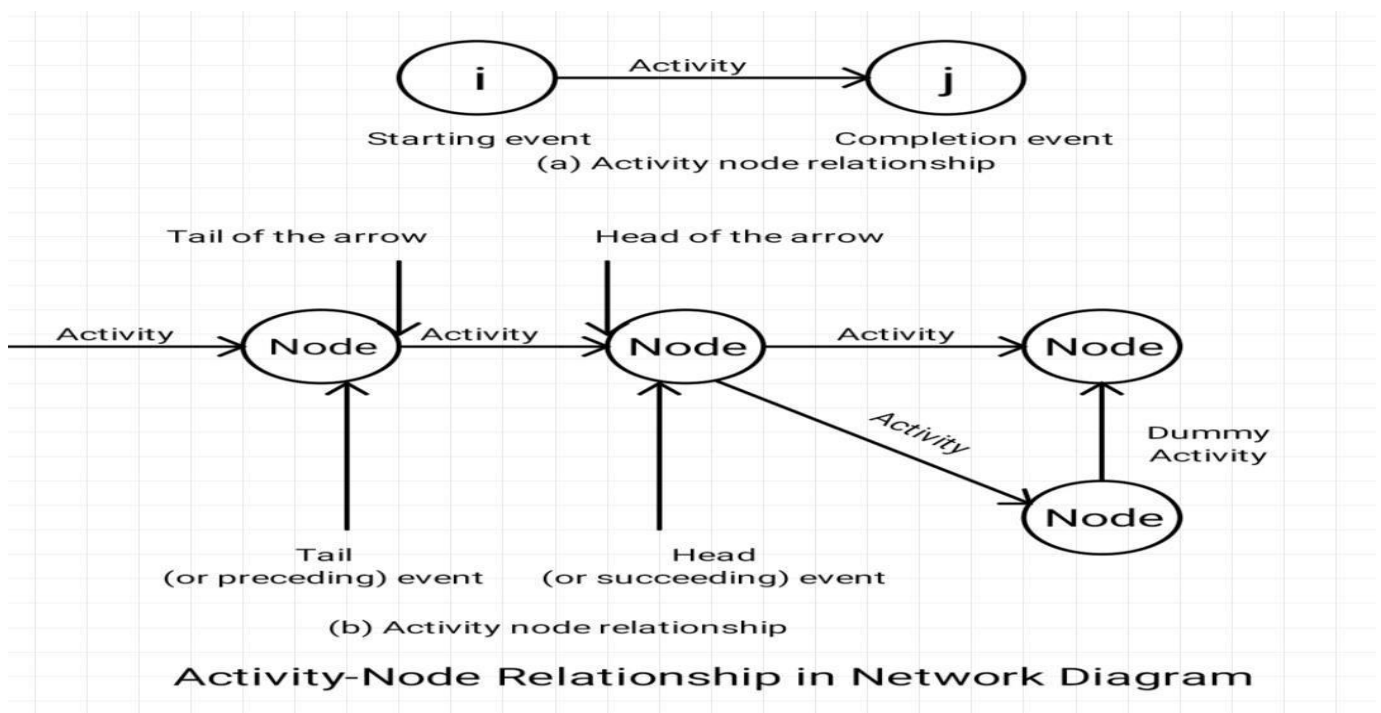
- **Merge Event** : An event which represents the joint completion of more than one activity is known as merge event .
- **Burst Event** : An event that represents the initiation (beginning) of more than one activity is known as burst event.



2. **Activities** : Activities in the network diagram represent project operations (or tasks) to be conducted. **Activity** is a distinct task that needs to be performed as part of a project.

The activities can be further classified into the following three categories:

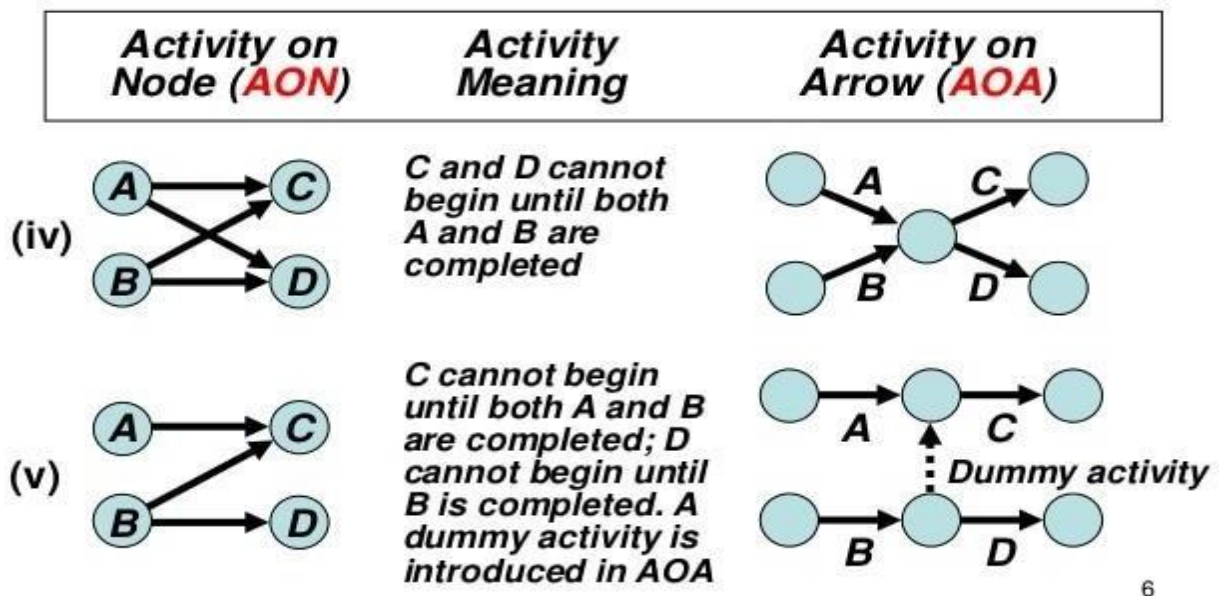
- **Predecessor Activity**: An activity which must be completed before one or more other activities start is known as predecessor activity.
- **Successor Activity**: An activity which starts immediately after one or more of other activities are completed is known as successor activity.
- **Dummy Activity**: An activity which does not consume either any resource and/or time is known as dummy activity. A dummy activity in the network is added only to establish the given precedence relationship among activities of the project. It is needed when (a) two or more parallel activities in a project have same head and tail events, or (b) two or more activities have some (but not all) of their immediate predecessor activities in common.



V. AON AND AOA NETWORKS

1. **Activity-on-Node (AON) network** : In this type of precedence network each node (or circle) represents a specific task while the arcs represent the ordering between tasks. AON network diagrams place the activities within the nodes, and the arrows are used to indicate sequencing requirements. Generally, these diagrams have no particular starting and ending nodes for the whole project. The lack of dummy activities in these diagrams always make them easier to draw and to interpret.
2. **Activity-on-Arrow (AOA) network** : In this type of precedence network at each end of the activity arrow is a node (or circle). These nodes represent points in time or instants, when an activity is starting or ending. The arrow itself represents the passage of time required for that activity to be performed.

A Comparison of AON and AOA Network Conventions



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VI. CRITICAL PATH ANALYSIS : Critical path is the longest path through the project network; the activities on the path are the critical activities, therefore any delay in their completion must be avoided to prevent delay in project completion.

VII. FLOAT (SLACK) OF AN ACTIVITY AND EVENT

The **float (slack) or free time** is the length of time in which a non-critical activity and/or an event can be delayed or extended without delaying the total project completion time.

- **Slack of an Event** : The *slack* (or *float*) of an event is the difference between its latest occurrence time (L_i) and its earliest occurrence time (E_i). That is: Event float = $L_i - E_i$. It is a measure of how long an event can be delayed without increasing the project completion time.

- **Slack of an Activity** : It is the amount of activity time that can increased or delayed without delaying project completion time. This float is calculated as the difference between the latest finish time and the earliest finish time for the activity.

There are three types of floats for each non-critical activity in a project.

(a) Total float: This is the length of time by which an activity can be delayed until all preceding activities are completed at their earliest possible time and all successor activities can be delayed until their latest permissible time.

$$\text{Total float (TF}_{ij}) = (L_j - E_i) - t_{ij} = LS_{ij} - ES_{ij} = LF_{ij} - EF_{ij}$$

(b) Free float: This is the length of time by which the completion time of any non-critical activity can be delayed without causing any delay in its immediate successor activities. The amount of free float time for a non-critical activity (i, j) is computed as follows:

$$\text{Free float (FF}_{ij}) = (E_j - E_i) - t_{ij} = \text{Min} \{ES_{ij}, \text{ for all immediate successors of activity } (i, j)\} - EF_{ij}$$

(c) Independent float: This is the length of time by which completion time of any non-critical activity (i, j) can be delayed without causing any delay in its predecessor or the successor activities. Independent float time for each non-critical activity is computed as follows:

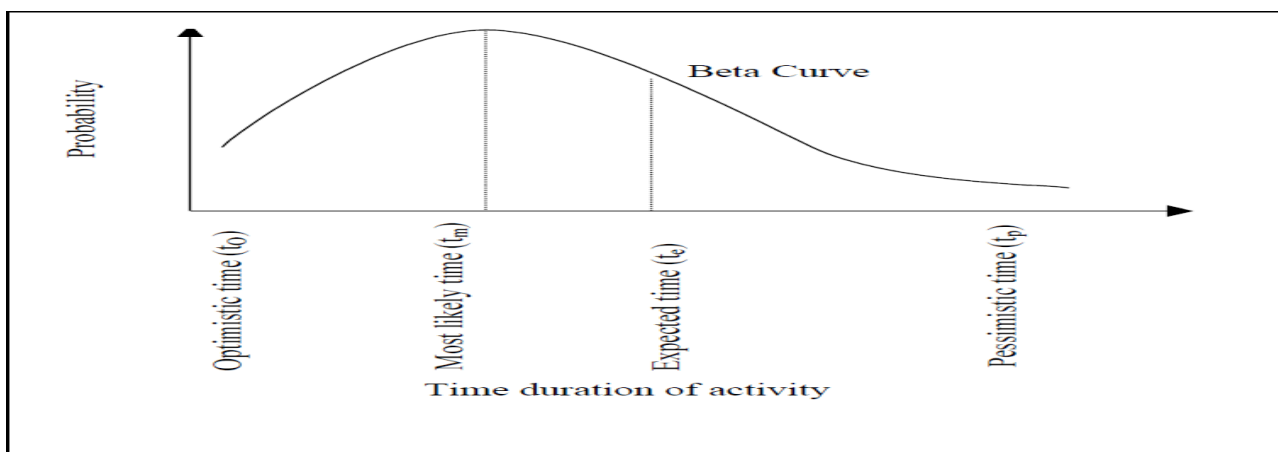
$$\text{Independent float (IF}_{ij}) = (E_j - L_i) - t_{ij} = \{ES_{ij} - LS_{ij}\} - t_{ij}$$

VIII. PERT (Program Evaluation and Review Technique)

PERT was developed to handle projects where the time duration for each activity is not known with certainty but is a random variable that is characterized by β (beta)-distribution. To estimate the parameters: mean and variance, of the β -distribution three time estimates for each activity are required to calculate its expected completion time.

The three-time estimates that are required are as under.

1. **Optimistic time (to or a)** : The shortest possible time (duration) in which an activity can be performed assuming that everything goes well.
2. **Pessimistic time (tp or b)** : The longest possible time required to perform an activity under extremely bad conditions. However, such conditions do not include natural calamities like earthquakes, flood, etc.
3. **Most likely time (tm or m)** : The time that would occur most often to complete an activity, if the activity was repeated under exactly the same conditions many times. Obviously, it is the completion time that would occur most frequently (i.e. model value).

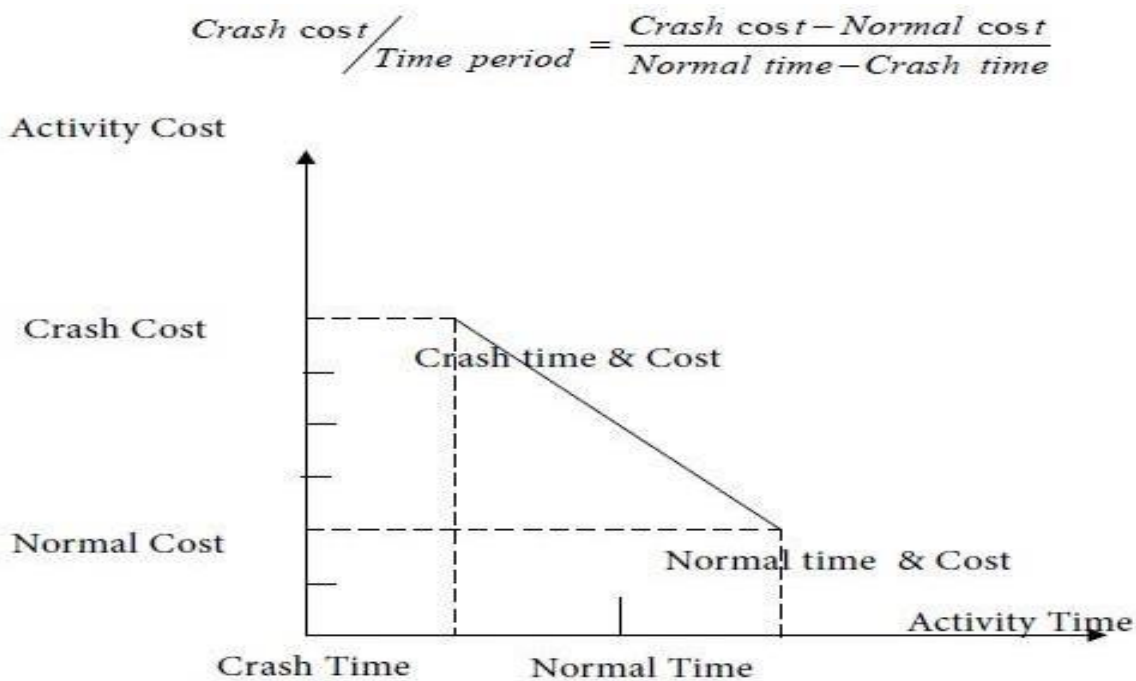


IX . PROJECT TIME-COST TRADE-OFF (PROJECT CRASHING)

The project completion time can be reduced by reducing (crashing) the normal completion time of critical activities. The crashing of normal completion time of critical activities will increase the total cost of the project. But, the decision-maker will always look for trade-off between the total cost of project and the total time required to complete it.

Crashing the project means crashing a number of activities to reduce the duration of the project, below its normal time.

Crashing of completion time of critical activities to reduce the project completion time would certainly require extra resources (cost). However, as shown in Fig. project cost increases when time is reduced. Similarly, the time increases while the project cost decreases. Obviously, an activity time may not be reduced beyond a certain limit and this limit is referred as *crash point*. Also, extending an activity time beyond normal point (cost efficient) may increase cost of executing that activity. For simplicity, the relationship between normal time and cost as well as crash time and cost for an activity is assumed to be linear. Thus, the crash cost per unit of time can be estimated by computing the relative change in cost (cost slope) per unit change in time.



TIME-COST TRADE-OFF PROCEDURE

Crashing an activity means taking special costly measures to reduce the duration of an activity below its normal time. The starting point for crashing is when all critical activities are completed with their normal time, and crashing is stopped when all critical activities are crashed.

The method of establishing time-cost trade-off for the completion of a project can be summarized as follows:

Step 1: Determine the normal project completion time and associated critical path.

Step 2: Identify critical activities and compute the cost slope for each of these by using the relationship

$$\text{Cost slope} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}}$$

$$\text{Normal time} - \text{Crash time}$$

The values of cost slope for critical activities indicate the direct extra cost required to execute an activity per unit of time.

- Step 3:** For reducing the total project completion time, identify and crash an activity time on the critical path with lowest cost slope value to the point where
- Another path in the network becomes critical, or
 - The activity has been crash to its lowest possible time.
- Step 4:** If the critical path under crashing is still critical, return to Step 3. However, if due to crashing of an activity time in Step 3, other path(s) in the network also become critical, then identify and crash the activity(s) on the critical path(s) with the minimum joint cost slope.
- Step 5:** Terminate the procedure when each critical activity has been crashed to its lowest possible time. Determine total project cost (indirect cost plus direct cost) corresponding to different project durations.

X. DIFFERENCE BETWEEN PERT AND CPM

Basic	PERT	CPM
Stands for	PERT stands for "Program Evaluation and Review Technique".	CPM stands for "Critical Path Method".
Model	It is a probabilistic model under which the result estimated in a manner of probability.	It is a deterministic model under which the result is ascertained in a manner of certainty.
Time	It deals with the activities of uncertain time.	It deals with the activities of precise well known time.
Jobs	It is used for onetime projects that involve activities of non-repetitive nature.	It is used for completing of projects that involve activities of repetitive nature.
Orientation	It is activity oriented in as much as its result is calculated on the basis of the activities.	It is even oriented, in as much as its results are calculated on the basis of the events.
Dummy Activities	It does not make use of dummy activities.	It makes use of dummy activities to represent the proper sequencing of the activities.
Cost	It has nothing to do with cost of a project.	It deals with the cost of a project schedules and their minimization.
Estimation	It finds out expected time of each activity on the basis of three types of estimates.	Its calculation is based on one type of time estimation that is precisely known.

XI. RULES FOR DRAWING NETWORK

Rules for Drawing Network

1. Each activity is represented by one and only one arrow in the network.
2. No two activities can be identified by the same end events
3. Precedence relationships among all activities must always be maintained.
4. Dummy activities can be used to maintain precedence relationships only when actually required. Their use should be minimized in the network diagram

XII. APPLICATION OF PERT / CPM

Applications of CPM / PERT these methods have been applied to a wide variety of problems in industries and have found acceptance even in government organizations.

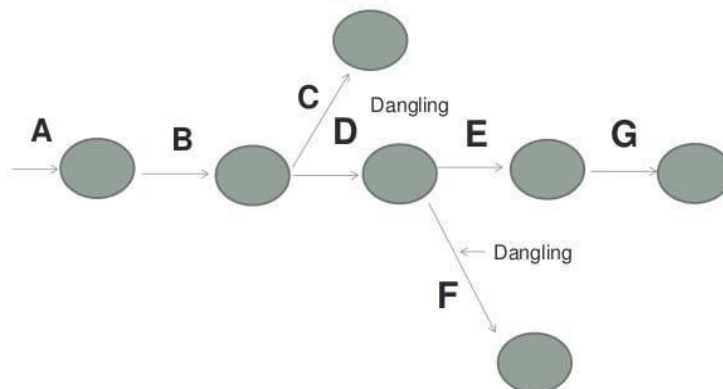
- Construction of a dam or a canal system in a region .
- Construction of a building or highway.
- Maintenance or overhaul of airplanes or oil refinery .
- Space flight.
- Cost control of a project using PERT / COST.
- Designing a prototype of a machine.
- Development of supersonic planes.

XIII. COMMON ERRORS IN DRAWING NETWORKS : The three types of errors are most commonly observed in drawing network diagrams

1. **Dangling:** To disconnect an activity before the completion of all activities in a network diagram is known as dangling. So the diagram is wrong and indicates the error of dangling.

Dangling

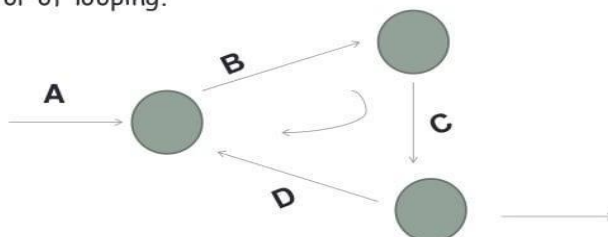
Whenever an activity is disconnected from the network it is called dangling error.



2. **Looping or Cycling** : Looping error is also known as cycling error in a network diagram. Drawing an endless loop in a network is known as error of looping as shown in the following figure.

Looping

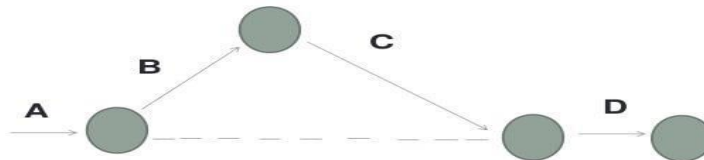
Looping error is also called as cycling error in a network diagram. Making an endless loop in a network is called as error of looping.



3. **Redundancy** : Unnecessarily inserting the dummy activity in network logic is known as the error of redundancy as shown in the following diagram.

Redundancy

When the dummy activity is introduced and it is not required, it is called redundancy errors.



XIV . PHASES OF PROJECT MANAGEMENT



1. **Initiating** : This process helps in the visualisation of what is to be accomplished. This is where the project is formally approved by the sponsor/client, initial scope defined, and stakeholders identified. This process is performed so that projects and programmes are not only approved by a sponsoring body, but also so that projects are aligned with the strategic objectives of the organisation. Where this is not performed, projects may be started and carried out haphazardly, with no real stated goal or objective.
2. **Planning** : This is a crucial process in project management. The planning process is at the heart of the project activity cycle, and gives guidance to stakeholders on where and how to undertake the project. The planning stage is where the project plans are documented, the project deliverables and requirements are defined, and the project schedule is created. It involves creating a set of plans to help guide your team through the implementation and closure phases of the project. The plans created during this phase will help the project team manage time, cost, quality, changes, risk and related issues.
3. **Executing** : This process is also known as the implementation phase, in which the plan designed in the previous phase of the project activity cycle is put into action. The intent of the execution phase of the project activity cycle is to bring about the project's expected results. Normally, this is the longest phase of the project management life cycle, where most resources are applied. During the project execution, the execution team utilises all the schedules,

procedures and templates that were prepared and anticipated during prior phases. Unexpected events and situations will inevitably be encountered, and the project manager and project team will have to deal with them as they arise.

4. **Monitoring and control :** This process oversees all the tasks and metrics needed to guarantee that the agreed and approved project that is undertaken is within scope, on time and within budget so that the project proceeds with minimum risk. This process involves comparing actual performance with planned performance and taking corrective action to yield the desired outcome when significant differences exist.
5. **Closing :** This is considered to be the last process of the project activity cycle. In this stage, the project is formally closed and then a report is produced to the project sponsor/client on the overall level of success of the completed project. The closing process involves handing over the deliverables to the sponsor/client, handing over documentation to the owners, cancelling supplier contracts, releasing staff and equipment, and informing stakeholders of the closure of the project.